

# DuPont Tedlar AI Lead Generation System

## 1. Solution Architecture

A **four-stage AI GTM workflow** that progressively identifies, scores and filters leads:



## 2. Data Processing Pipeline

Details found in [prompts.py](#).

| STAGE 1: EVENT DISCOVERY                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                 |
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| <div><b>Step 1.1: Discover</b><br/>Tool: Claude WebSearch<ul style="list-style-type: none"><li>• Sign industry shows</li><li>• Graphics/printing</li><li>• Vehicle wrap events</li><li>• Architectural</li></ul><b>Output: 22 events</b></div> | <div><b>Step 1.2: Score &amp; Filter</b><br/>Tool: Claude LLM<br/><br/>Scoring (each 0-10):<ul style="list-style-type: none"><li>• Industry Alignment (50%)</li><li>• Exhibitor Quality (30%)</li><li>• Scale &amp; Timing (20%)</li></ul><b>Tier 1 (8-10): 11 → Proceed</b><br/><b>Tier 2 (5-7.9): 9 → If budget</b><br/><b>Tier 3 (&lt;5): 2 → Skip</b></div> |
| <div><b>Step 1.3: Company Discovery (for Tier 1 events)</b><br/>Tool: Claude WebSearch → Output: 37 unique companies with booth/website info</div>                                                                                             |                                                                                                                                                                                                                                                                                                                                                                 |

| STAGE 2: COMPANY QUALIFICATION                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Step 2.1: Deep Research</b><br>Tool: Claude WebSearch<br><br>For each company: <ul style="list-style-type: none"><li>• Company overview &amp; history</li><li>• Products/services offered</li><li>• Revenue &amp; employee count</li><li>• Recent news &amp; initiatives</li><li>• Trade show presence</li></ul> <b>Output: research.json</b> | <b>Step 2.2: ICP Scoring</b><br>Tool: Claude LLM<br><br>Scoring Categories: <ul style="list-style-type: none"><li>• Industry Fit (30%)</li><li>• Size/Revenue (25%)</li><li>• Strategic Relevance (20%)</li><li>• Market Activity (25%)</li></ul> <b>Tier 1 (85+): 5 companies</b><br><b>Tier 2 (70-84): 2 companies</b><br><b>Tier 3 (55-69): 1 company</b> |
| <b>Output: 8 qualified companies → proceed to Stage 3</b><br>Each with: scoring.json (scores + rationale per category)                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                              |

| STAGE 3: TARGET ROLE IDENTIFICATION                                                                                                                                                                   |                                                                                                                                                                                                                            |
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| ⚡ Auto-triggers when Stage 2 scores company as Tier 1 or Tier 2                                                                                                                                       |                                                                                                                                                                                                                            |
| <b>Input</b><br>From Stage 2: <ul style="list-style-type: none"><li>• Company research data</li><li>• ICP scoring breakdown</li><li>• Identified pain points</li><li>• Strategic priorities</li></ul> | <b>Process</b><br>Tool: Claude LLM <ul style="list-style-type: none"><li>• Analyze org structure</li><li>• Identify decision-makers</li><li>• Rank by purchasing authority</li><li>• Generate outreach rationale</li></ul> |
| <b>Output per company: 3 prioritized target roles</b><br>Each with: title, rationale, LinkedIn Sales Navigator search URL                                                                             |                                                                                                                                                                                                                            |

| STAGE 4: OUTREACH GENERATION                                                                                                                                              |                                                                                                                                                                                                       |                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Input</b> <ul style="list-style-type: none"><li>• Company research</li><li>• ICP scores</li><li>• Target role</li><li>• Role rationale</li><li>• Pain points</li></ul> | <b>Step 4.1: Analysis</b><br>Tool: Claude LLM <ul style="list-style-type: none"><li>• Find personal hooks</li><li>• Determine angle</li><li>• Match pain points</li><li>• Identify triggers</li></ul> | <b>Step 4.2: Generate</b><br>Tool: Claude LLM <ul style="list-style-type: none"><li>• 80-150 word email</li><li>• Consultative tone</li><li>• Company-specific</li><li>• Soft CTA</li></ul> |
| <b>Output: Personalized outreach message</b><br>Subject line + email body with company-specific hooks and value proposition                                               |                                                                                                                                                                                                       |                                                                                                                                                                                             |

3. AI Workflow Orchestration

Pipeline is orchestrated by `main.py` that relies on event triggers with:

1. **Priority order:** Later stages processed first such that companies closer to completion are processed through the pipeline before starting new ones.
2. **Automatic triggers:** Callbacks connect stages (when Stage 2 completes, Stage 3 starts automatically for qualified companies).
3. **Rate limiting:** Shared limiter tracks API tokens (30k/minute) across all stages.
4. **Resume capability:** Each stage produces JSON artifacts, enabling resume from interruption and preventing duplicate reruns.
5. **Tiered qualification:** Scoring thresholds before proceeding to next stage ensures resources focus on high-potential leads. Low-scoring companies filtered early.

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## 4. Implementation Results

### 4.1 Data Outputs

Each stage produces JSON artifacts stored in `data/`:

```
data/
├── events/
│   ├── discovered_events.json      # 22 raw events from Stage 1.1
│   ├── scored_events.json          # Tiered events from Stage 1.2
│   ├── discovered_companies.json   # 37 companies from Stage 1.3
│   └── companies/
│       └── {EventName}.json        # Companies per event
├── companies
│   └── all_qualified_companies.json # All discovered companies scored
against ICP
│   └── {CompanyName}/
│       ├── research.json           # Stage 2.1 deep research
│       ├── scoring.json            # Stage 2.2 ICP scores
│       └── target_roles.json        # Stage 3 decision-makers
└── contacts/
    ├── role_identification_results.json
    ├── linkedin_sales_nav_searches.json
    ├── clay_push_results.json
    └── {CompanyName}/
        ├── {ContactName}_analysis.json # Stage 4.1 hooks/angles
        └── {ContactName}_outreach.json  # Stage 4.2 email draft
```

## 5. Limitations & Future Work

### Current Limitations:

- Contact discovery generates custom LinkedIn Sales Navigator URL, but workflow doesn't route the actual LinkedIn profiles discovered back for the last stage (ie. personalized outreach to each profile)
- Rate limits require ~60s min. delays between API calls

### Potential Enhancements:

- Add email verification step before outreach
- CRM integration for direct lead export